

Sharayu Rasal

New York, NY | ✉ srr10019@nyu.edu | ☎ +1 (929) 663-1867 | 🌐 sharayu.dev | in LinkedIn | 🐙 GitHub

Education

- New York University**, Master of Science (M.S.) in Computer Science (**GPA: 3.78/4.0**) Sep 2024 – May 2026
- **Coursework:** Cloud Computing, Design & Analysis of Algorithms, Software Engineering, Machine Learning, Computer Networks
- Pune University**, B.E. in Computer Science (Honors in Data Science) (**CGPA: 8.15/10.0**) Aug 2020 – May 2024
- **Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Distributed Systems, DBMS

Skills

Languages: Python, C++, TypeScript, JavaScript, SQL

Distributed Systems & Cloud: AWS (S3, Lambda, Kinesis, API Gateway, SQS, DynamoDB, ECS, OpenSearch, CloudFormation, IAM, CloudWatch, EC2, RDS), Kafka, Docker, Kubernetes

Data Engineering & Storage: MongoDB, PostgreSQL, DynamoDB (Single Table Design, Indexing), Redis (Caching Strategies), Elasticsearch

System Design & Resiliency: Event-Driven Architecture, Async Patterns, Idempotency, DLQ, Fault Tolerance, Microservices

Backend Frameworks: Django, FastAPI, Node.js, GraphQL, REST APIs, JWT Authentication

Tools & Observability: Git, Linux, CI/CD (GitHub Actions), Prometheus, CloudWatch

Professional Experience

- Research & Development Intern, Bharat Forge Ltd (KCTI)** Jul 2023 – Nov 2023
- Led development of backend components of a real-time computer vision pipeline built using **C++ and Python**, optimizing data flow to reduce end-to-end latency by **18%** under constrained compute and network conditions.
 - Proposed and implemented reliable **data transmission and buffering mechanisms** to stabilize throughput and reduce performance variance across **unstable bandwidth environments**.
 - Collaborated with **cross-functional teams** to integrate **backend modules**, formalize **API contracts**, and support **production deployment**, improving integration reliability and release readiness.
- Software Engineering Intern, Posit Source Technologies** Mar 2023 – Jun 2023
- Launched and maintained **scalable backend services and ETL pipelines** in Python to process high-volume logistics data.
 - Developed **RESTful APIs** and internal tooling to support real-time analytics, reducing manual reporting effort by **60%**.
 - Optimized **Redis-based caching** and task scheduling to improve service latency and throughput, reducing delivery lead time by **30%** under peak load.
- Technical Product Intern – Web, Oniv Beverages Pvt Ltd** Dec 2023 – May 2024
- Engineered a **React-based consumer-facing web app** with **client-side event tracking** integrated into AWS ingestion pipelines, contributing to a **30% increase in organic traffic**.

Selected Projects

- Research Paper RAG System (Serverless Semantic Search Platform)** 🔄
- Built a **serverless, event-driven retrieval platform** on AWS Lambda, DynamoDB, and S3 to enable **low-latency semantic search** over **1k+ documents** under concurrent query load.
 - Designed a **fault-tolerant ingestion pipeline** with SQS to decouple upload, extraction, embedding generation, and indexing for **high-throughput processing**.
 - Implemented secure, **low-latency Q&A APIs** with Cognito/JWT, ensuring per-user isolation, HTTPS-only access, and encryption at REST.
- SmartCache AI (Distributed Content Orchestration Platform)** 🔄
- Engineered a **distributed background processing system** coordinating **asynchronous workers** for content discovery and delivery under **high-throughput workloads**.
 - Built **real-time state synchronization** (WebSockets/Redis) and **event-driven ingestion pipeline** (Celery) on ECS with round-robin agent coordination (AutoGen), supporting **20–50 concurrent requests** with stable latency and improved system reliability under load.
- Aegis (Real-Time Incident Management Platform)** 🔄
- Built and operated **resilient, event-driven platform** using AWS Lambda and SQS, reliably processing **1000+ incidents per hour** while ensuring high availability for critical business workflows.
 - Architected an **extensible incident correlation engine** using **SHA-256 fingerprinting** to reduce alert noise and improve operational focus.
 - Composed and shipped **well-tested REST APIs** for real-time incident retrieval, maintaining **sub-100ms response times**.

Open Source Contributions

- Responsible-Alliance-Protocol (Kubernetes / Security)** 🔄
- Owned and shipped production-grade **Kubernetes manifests** for an OPA-based policy service with **namespace isolation, quotas/limits, ConfigMaps**, and default-deny **NetworkPolicies**; enabled **HA** (merged).
- Developer-solutions-lab (AWS / TypeScript)** 🔄
- Built an **offline AWS region detection diagnostic** for on-prem readiness with **zero network/credential dependency** (merged).

Leadership

- **Global Ambassador, New York University:** Led outreach and onboarding coordination for incoming graduate students, improving early engagement and easing transition into academic and research communities.
- **Core Team Member, Google Developer Student Club (Cloud):** Organized and led cloud/APIs workshops for 300+ students, increasing adoption of modern development practices and fostering peer learning.